SLIDE 1 • TITLE SLIDE

RiskEngine: AI/NLP Financial Risk Quantification

Transforming Unstructured News & Social Streams into Actionable Quantitative Risk Signals

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Competition: S&P Global & Crisil Campus Hackathon 2026

Core Implementations: Hybrid NLP Pipeline + Module A (Index Rebalancing) + Module B (Stress Testing)

Test Suite: 165 passing automated unit & integration tests

Repository: Complete end-to-end self-contained prototype (FastAPI backend + React 18 frontend)

SLIDE 2 • PROBLEM & APPROACH

Problem Statement & Strategic Approach

Moving Beyond Naive Sentiment Scores to Institutional Financial Risk Management

The Industry Dilemma: Financial institutions process millions of unstructured news and social items daily, yet naive NLP pipelines produce isolated sentiment polarities without financial risk context.

Key Shortcomings of Naive Approaches:

- Generic sentiment ignores event severity, source credibility, repetition vs. novelty, and time decay.
- NLP models rarely interface with actual portfolio risk constraints or stress-testing frameworks.

Our Multi-Tiered Solution Approach:

- Ingestion & Gating: Dual-stream ingestion (news + social) with SHA-1 deduplication and source reliability weighting.
- Multi-Factor Risk Quantification: Ordinal 1.0–10.0 impact index combining FinBERT/lexicon sentiment, 14-class event taxonomy, novelty decay, and recency half-life.
- Dual Downstream Execution: Signals actuate Module A (Tactical Index Rebalancer) and Module B (Wholesale Portfolio Stress Tester).

SLIDE 3 • SYSTEM DESIGN

System Architecture & Data Flow

Modular, End-to-End Pipeline from Unstructured Ingestion to Risk Execution

1. Ingestion Layer: NewsAPI, Twitter cashtags (*AAPL*, TSLA), GDELT; SHA-1 fingerprinting suppresses duplicate wire feeds.

   Source trust priors: Reuters/Bloomberg 0.95, NewsAPI 0.75, GDELT 0.70, Kaggle-financial 0.80, Tweets 0.40.

2. NLP Extraction: Ticker NER via cashtag + company-alias matching; FinBERT financial sentiment (lexicon fallback); 14-class event taxonomy.

   Event classes: Geopolitical, Macroeconomic, Credit Event, M&A, Earnings, Product Launch, Regulatory, Legal,

   Supply Chain, Management Change, Cybersecurity, Market Shock, Commodity Shock, Other.

3. Multi-Factor Risk Signal Engine:

   raw = (0.60 * severity + 0.40 * |sentiment|) * f_conf * f_rel * f_mkt * f_nov * f_rec

   impact = clamp(1.0 + 9.0 * raw^0.70, 1, 10) — each f_x ∈ [floor, 1.0]

4. Module A (Portfolio Rebalancer): 20-constituent S&P 100 proxy, live yfinance prices, L1 turnover cap.

5. Module B (Stress Testing): ~$89.92M 20-position banking book; auto-triggers when adverse impact ≥ 7.0.

SLIDE 4 • IMPLEMENTATION HIGHLIGHTS

Implementation Highlights & Tech Choices

Built for Reliability, Transparency, and Quantitative Credibility

FastAPI Backend & Strict Pydantic Schemas: Validated data contracts across all pipeline stages; interactive Swagger docs auto-generated.

FinBERT + Lexicon Hybrid NLP: Transformer-based sentiment when transformers/torch are installed; automatic lexicon fallback otherwise. Both paths produce identical interface contracts.

Live Market Data (yfinance): Real-time prices, 30-day volatility, and historical returns for the 20-stock S&P 100 universe; transparent simulated fallback when offline.

Institutional L1 Turnover Budget (Module A): Prevents unrealistic churn by enforcing L1 turnover ≤ 30% (default); configurable per request.

Multi-Component Stress Transmission (Module B): Equity shock × sector amplifier + duration × Δy (DV01) + credit spread DV01 + stressed EL + FX + commodity vol.

Full React 18 + TypeScript + Recharts Dashboard: Dark-mode institutional workstation with interactive sandboxes, scenario selectors, and multi-dimensional loss attribution charts.

SLIDE 5 • KEY RESULTS

Key Quantitative Results & Verification

Concrete Outputs, Verified Against Code

Test Suite: 165 automated unit & integration tests — 100% passing — covering scoring invariants, NLP correctness, portfolio math, and stress-test accounting.

Module A (Tactical Rebalancing):

- Rebalances 20-stock S&P 100 proxy using sentiment tilt: $w_i \propto w_{base,i} \times (1 + 0.40 \times sentiment_i) \times vol_adj_i$.
- Turnover is capped at 30% total (L1 norm); min/max weight bounds [2%, 12%] prevent concentration.
- Backtest computes cumulative returns vs. equal-weight and cap-weight benchmarks over historical windows.

Module B (Wholesale Portfolio Stress Testing):

- Stress-tested across a synthetic ~$89.92M 20-position banking book (5 corporate loans, 3 gov bonds, 4 corporate bonds, 4 equities, 3 derivatives, 1 cash).
- Illustrative scenario losses: e.g., Geopolitical Escalation (severity 8.5) → position-level shock breakdown.
- Granular attribution by asset class, sector, geography, and credit rating band (AAA → High-Yield).

Ticker Disambiguation: Test regressions reject 'apple pie', 'amazon rainforest', 'BA flight delayed' as false positives.

SLIDE 6 • DOMAIN IMPACT

Domain & Business Impact

Empowering Credit Analysts, Risk Officers, and Portfolio Managers

For Credit Rating Agencies & Risk Analysts (Crisil / S&P Global):

- Replaces manual news monitoring with automated 14-class event classification and early-warning triggers.
- Breaking headlines propagate into wholesale book stress tests without waiting for quarterly reviews.

For Asset Managers & Index Funds:

- Sentiment tilting allows capturing momentum or hedging against adverse corporate announcements within strict turnover envelopes.

For Regulatory Compliance & Governance:

- Full auditability: Every weight shift or stress dollar loss traces back to raw source text, SHA-1 fingerprint, and mathematical factor decomposition.
- Transparent simulated data labels: every synthetic record is marked simulated=True in the API response.

SLIDE 7 • LIMITATIONS & NEXT STEPS

Limitations, Assumptions & Future Roadmap

Honest Engineering Disclosures & Scalability Pathways

Current Assumptions & Scope:

- 20-stock S&P 100 proxy — not a full production index; chosen to stay within API quota and latency budgets.
- Impact scores are ordinal severity indices (1.0–10.0), NOT price forecasts or calibrated probabilities.
- Module B banking positions are synthetic — illustrative of institutional structure, not real trades.
- FinBERT requires torch + transformers; default install uses lexicon fallback (clearly disclosed at /health).
- Stress correlations between shocks are not modeled (shocks applied independently per position).

Future Roadmap:

- Scale to full S&P 500 / Russell 1000 with real-time WebSocket streaming from Bloomberg / FactSet.
- Multi-lingual transformer models for international sovereign risk monitoring.
- Quadratic programming optimizer (Markowitz mean-variance with covariance shrinkage) for Module A.
- Correlation matrix between shock factors for Module B (Cholesky decomposition).